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Can Sad Music Really Make You Sad? Indirect Measures of Affective States Induced by Music and Autobiographical Memories

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The present study addressed music's disputed ability to induce genuine sadness in listeners by investigating whether listening to sad music can induce sadness-related effects on memory and judgment. Related aims were to explore how the different mechanisms of music-induced emotions are involved in sadness induced by familiar, self-selected music and unfamiliar, experimenter-selected music, and whether the susceptibility to music-induced sadness is associated with trait empathy. One hundred twenty participants were randomly assigned into four conditions with different tasks: listening to unfamiliar sad or neutral music, or to self-selected sad music, or recalling a sad autobiographical event and writing about it. The induced affective states were measured indirectly using a word recall task and a judgment task where participants rated the emotions expressed by pictures depicting facial expressions. The results indicate that listening to sad music can indeed induce changes in emotion-related memory and judgment. However, this effect depends, to some extent, on the music's relevance to the listener, as well as on the personality attributes of the listener. Trait empathy contributed to the susceptibility to sadness induced by unfamiliar music, while autobiographical memories contributed to sadness induced by self-selected music.

Keywords: music-induced emotion, sadness, empathy

It is a commonly accepted notion in the Western culture that music has the ability to both express emotions and induce emotional responses in the listener (see, e.g., Juslin & Sloboda, 2001). However, researchers in the field of music and emotions disagree on what kinds of emotions music can induce in listeners (e.g., Juslin & Laukka, 2004; Konečni, 2008), and whether the responses evoked by music are actually real emotions at all (e.g., Kivy, 1989, 1990; Scherer & Zentner, 2001; Konečni, 2008; Juslin & Västfjäll, 2008). Particularly problematic is the case of sadness, as many people report enjoying sad music despite its negative emotional tone (Garrido & Schubert, 2011; Huron, 2011; Vuoskoski, Thompson, McIlwain, & Eerola, in press). Although people often report experiencing sadness in response to music (Juslin & Laukka, 2004; Juslin, Liljeström, Laukka, Västfäll, & Lundqvist, 2011; Vuoskoski et al., in press), some researchers claim that these people are in fact mistaken about the quality of their experience, and that they confuse their own emotional experience with the emotion expressed by the music (see, e.g., Kivy, 1989; Konečni, 2008). The rationale behind this claim is that since music cannot have any real-life consequences for the listeners' goals or desires, it cannot evoke "real" emotions such as sadness—unless the music triggers memories of emotional events through association (Kivy, 1989, 1990).

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There has been a limited amount of empirical attempts to investigate music's ability to induce genuine emotional responses in listeners (for a review, see Konečni, 2008). As self-reports may be vulnerable to demand characteristics (Orne, 1962), and as some researchers claim that listeners may easily confuse emotional responses to music with emotions expressed by the music (Kivy, 1989, 1990), most of these studies have focused on physiological responses to music expressing different emotions—including sadness. Some studies have reported that sad music can elicit physiological reactions distinguishable from reactions elicited by certain other types of music (e.g., Nyklícek, Thayer, & Van Doornen, 1997; Krumhansl, 1997; Lundqvist, Carlsson, Hilmersson, & Juslin, 2009), while others have found that physiological reactions to music are largely accounted for by musical features such as tempo (Etzel, Johnsen, Dickerson, Tranel, & Adolphs, 2006; Gomez & Danuser, 2007). Therefore, it is difficult to say what the relative contributions of the acoustical and emotional aspects of music to the reported physiological reactions are. Despite the limitations related to self-reports (especially in the context of music-induced emotions), there have also been self-report studies that have addressed music's ability to induce "real" emotions: Using well-known pieces from the Western classical music repertoire, Konečni, Brown, and Wanic (2008) compared the emotional states induced by happy, neutral, and sad music and the recollection of autobiographical events. They found that sad music evoked lower ratings of experienced sadness than the recollection of sad autobiographical events, but happy music lead to ratings that were comparable to the ratings evoked by the recollection of happy events. However, it is premature to conclude that sadness is less intensely induced by music than happiness, since the study design (i.e., only one sad and one happy piece used as stimuli) leaves alternative explanations open.

In addition to a host of studies from the fields of music therapy, music psychology, and music education, some evidence for music's ability to induce affective states comes from studies utilizing music listening (with or without instructions to deliberately change one's mood with the help of music) as a method of "mood manipulation" (for a review, see Västfjäll, 2002). Despite the fact that a number of studies have used music listening to manipulate participants' mood states (see, e.g., Westermann, Spies, Stahl, & Hesse, 1996; Västfjäll, 2002), Juslin and Västfjäll (2008) argue that the affective responses evoked by music are better characterized as emotions rather than moods, since the listeners' responses are focused on an object (the music), their duration is relatively short (ca. 5–40 minutes; see Västfjäll, 2002), and they involve components that are commonly associated with emotions rather than moods. One issue that may have contributed to the debatable use of music listening as "mood-manipulation" is that affect researchers sometimes use the terms *mood* and *emotion* interchangeably—partly because of a lack of empirical evidence regarding emotion–mood distinctions (Beedie, Terry, & Lane, 2005). Furthermore, it is also unclear whether emotions and moods are two distinct phenomena, or whether they actually represent the opposite ends of a single continuum (Beedie et al., 2005). The present study adopts the view that, although the affective responses to music may be better described as emotions rather than moods, clear distinctions cannot be made between the two, as affective processing may be better described as a continuous process (see, e.g., Sloboda & Juslin, 2010).

Some of the so-called mood-manipulation studies have used indirect measures of affective states (for a review, see Västfjäll, 2010), which are based on the premise that affective states like emotions and moods are accompanied by changes in information processing and behavior (see, e.g., Bower, 1981; Russell, 2003), and that music-induced emotional states last longer than the duration of the music piece listened to (see, e.g., Västfjäll, 2002). These indirect measures are relatively free from demand characteristics, and do not rely on the participants' conscious interpretations of their own internal processes (which may not always be accessible on a conscious level; see, e.g., Nisbett & Wilson, 1977). Examples of indirect measures that have been used to study emotional states induced by music include word recall (Clark & Teasdale, 1985), autobiographical recall (Parrot & Sabini, 1990), count time (Clark & Teasdale, 1985), distance estimation (Kenealy, 1988), and emotional picture judgment tasks (Bouhuys, Bloem, & Groothuis, 1995).

Although musical mood-induction studies have shown that music can effectively induce different emotional states in participants (for reviews, see Westermann et al., 1995; Västfjäll, 2002), most of the studies have focused on comparing the effects of happy and sad music (e.g., Clark & Teasdale, 1985; Kenealy, 1988; Parrot & Sabini, 1990), which differ both in arousal and valence. As music's ability to induce happiness is less controversial than its ability to induce sadness (e.g., Konečni et al., 2008), the differentiation between the effects induced by happy and sad music is not enough to suggest that sad music can indeed induce sadness. Perhaps one of the most interesting findings regarding musical mood-induction studies concerns the effectiveness of different types of music: Carter, Wilson, Lawson, and Bulik (1995) found that music selected by the participants themselves was more effective in inducing the intended emotional states than music chosen by the experimenters. This finding highlights the personal relevance of music-induced emotional states, and raises questions about the role of autobiographical memories (or *episodic memories*, see Juslin & Västfjäll, 2008) and subjective associations in music-induced emotions.

Juslin and Västfjäll (2008) have proposed different mechanisms through which music can induce emotions in listeners. These include *brain stem reflexes*, *evaluative conditioning*, *emotional contagion*, *visual imagery*, *episodic memory*, and *musical expectancy*. Apart from a few explorative studies (e.g., Juslin, Liljeström, Västfjäll, Barradas, & Silva, 2008; Juslin et al., 2011), little is known about the prevalence of these suggested mechanisms with regard to different emotions. It may be that some of these suggested mechanisms are more effective than others in inducing sadness, while other mechanisms (e.g., emotional contagion) may interact with personality attributes such as trait empathy. Trait empathy can be defined as an individual's responsivity to the observed experiences of another, involving both perspective-taking capabilities or tendencies, and emotional reactivity (Davis, 1980). Trait empathy has previously been linked with emotional contagion in general (e.g., Doherty, 1997), and it may also contribute to susceptibility to music-induced emotions (see, e.g., Scherer & Zentner, 2001; Vuoskoski & Eerola, 2011).

The present study addresses music's disputed ability to induce genuine sadness in listeners by investigating whether listening to sad music can induce sadness-related effects on memory and judgment. Autobiographical recall has been shown to be one of the most effective ways to induce emotional states (see, e.g., Brewer, Doughtie, & Lubin, 1980; Baker & Gutterfriend, 1993; Jallais & Gilet, 2010), and should thus serve as a valid point of comparison for musically induced emotional states. The memory and judgment effects induced by sad music and sad autobiographical recall will be directly compared to the effects induced by neutral music (rather than happy music). This comparison will naturally be subtler than a comparison with the effects of happy music, but will form a more controlled setting in terms of both valence and arousal. A related aim was to explore how some of the proposed mechanisms of music-induced emotions (namely emotional contagion and episodic memories; Juslin & Västfjäll, 2008) are involved in sadness induced by self-selected music, and unfamiliar, experimenter-selected music. A possible explanation behind the reported superiority of self-selected music in inducing emotional states (Carter et al., 1995) may be that different mechanisms contribute to the emotional states induced by self-selected and unfamiliar music. It may also be that there are individual differences in the susceptibility to music-induced emotions, as mechanisms such as emotional contagion may interact with a person's inherent tendency to experience empathy. Thus, it was also investigated whether the susceptibility to sadness induced by unfamiliar music is associated with trait empathy. The induced emotional states were measured indirectly using a word recall task and a judgment task where participants rated the emotions expressed by facial expression pictures.

According to the associative network theory of affect (Bower, 1981), emotional states bias the interpretation of emotional material—especially ambiguous material—in affect-congruent directions. Previous studies have shown that induced affective states lead to affect-congruent judgment biases in the evaluation of pleasant, unpleasant, and ambiguous slides (Isen & Shalker, 1982)

and drawn faces with clear and ambiguous emotional expressions (Bouhuys et al., 1995), for example. In other words, people experiencing a sad emotional state would interpret emotional stimuli as sadder and less happy than people experiencing a neutral or happy emotional state (for a review, see Rusting, 1998).

The associative network theory (Bower, 1981) also postulates that people remember more easily material (such as words or memories) that is affectively congruent with their current emotional state. However, empirical studies have reported inconsistent findings: affect-congruent memory effects, no memory effects, and affect-incongruent memory effects (for reviews, see Matt, Vásquez, & Campbell, 1992; Rusting, 1998). Isen (1985) has suggested that affect-incongruent memory effects stem from participants' instinctive attempts to regulate their moods and emotions. In fact, Parrot and Sabini (1990) demonstrated in a series of experiments that affect-incongruent memory effects emerge in naturalistic conditions where participants are unaware of the relevance of their affective states to the experiment. In contrast, affect-congruent memory effects were observed when the participants were instructed to deliberately change their moods, thus possibly inhibiting their mood-regulation tendencies.

As music and emotion scholars seem to disagree in particular on whether unfamiliar, instrumental music can induce genuine sadness in listeners, the following competing hypotheses were formulated for direct testing:

> *Hypothesis 1A:* Both unfamiliar, instrumental sad music, and self-selected sad music—as well as sad autobiographical recall—will induce memory and judgment effects typically associated with experienced sadness, while neutral music will induce no effects.
>
> *Hypothesis 1B:* Only self-selected sad music and sad autobiographical recall will be able to induce memory and judgment effects, while unfamiliar, instrumental sad music, and neutral music will not induce memory or judgment effects.

In addition, it was hypothesized (Hypothesis 2) that trait empathy would contribute to the susceptibility to music-induced sadness—especially in the case of experimenter-selected, unfamiliar sad music.

Pilot Experiment

Stimulus Selection

The experimenter-selected, unfamiliar instrumental sad music had to meet the following criteria: 1. It conveys sadness in a clear manner, 2. The emotion conveyed does not change or vary during the course of the excerpt, 3. It is unfamiliar to most participants, and 4. The duration of the excerpt is at least 8 minutes (in order to induce a sad emotional state that would last for the duration of the indirect measures; for a review, see Västfjäll, 2002). After listening to several film soundtracks, an excerpt (duration 8 Minutes 35 seconds; starting at 2:25) from the piece *Discovery of the Camp* (*Band of Brothers* soundtrack; Kamen, 2001, track 17) was selected as the unfamiliar sad music. Furthermore, the first movement (*De l'aube à midi sur la mer*; duration 8 Minutes 35 seconds) of the orchestral work *La Mer* by Claude Debussy (used as neutral music previously by Martin & Metha, 1997) was chosen as the experimenter-selected neutral music. In addition, a piece of elating music (an 8-min excerpt from the piece *Band of Brothers Suite Two; Band of Brothers* soundtrack; Kamen, 2001, track 3) was chosen to bring the participants in the autobiographical recall condition to a more positive emotional state after the experiment.

Participants

The participants of the pilot experiment were 20 university students (from different faculties) aged 21–34 years (mean 25.3, $SD = 3.6$, 65% males).

Procedure

In order to confirm that the three excerpts conveyed the intended emotions, each excerpt was split into 14–15 shorter segments (approx. 35 seconds each), which were tested in a listening experiment along with two angry, two fearful, and two sad excerpts taken from a set of 110 film music excerpts (excerpts 001, 002, 011, 012, 031, and 032; Eerola & Vuoskoski, 2011). The participants were instructed to rate the emotions expressed by the excerpts using five discrete emotion scales (happiness, sadness, anger, fear, and tenderness) and two bipolar scales (valence and arousal). The five discrete emotions were selected on the bases that they are commonly expressed by music (see Juslin & Laukka, 2004) and cover all the quadrants of the two-dimensional valence-arousal space. All emotion ratings were done on scales from 1 to 7. In addition, the participants rated how familiar the excerpts sounded (1 = *unfamiliar*, 2 = *somewhat familiar*, and 3 = *very familiar*).

Results

The mean familiarity ratings for the sad, neutral, and elating excerpts were 1.14 ($SD = 0.36$), 1.06 ($SD = 0.24$), and 1.26 ($SD = 0.52$), respectively, confirming that the vast majority of participants found the excerpts unfamiliar. A repeated-measures ANOVA confirmed that the 14 segments of the experimenter-selected sad music expressed sadness more than any other emotion, $F(4, 76) = 75.4$, $p < .001$, η^2_G (Generalized Eta Squared; Bakeman, 2005) = .69. All post hoc comparisons of means (Tukey) between sadness and other emotions (happiness, tenderness, fear, and anger) were significant at the $p < .001$ level. The mean sadness rating for experimenter-selected sad music was 4.75 ($SD = 1.55$), while the mean sadness rating for a highly sad control stimulus (excerpt No. 031; Eerola & Vuoskoski, 2011) was 4.90 ($SD = 1.65$). For each of the 14 segments, sadness was the highest rated emotion, with the mean rating varying between 3.80 and 5.40. The mean valence and arousal ratings for the neutral excerpt were 4.01 ($SD = 1.27$) and 4.50 ($SD = 1.37$), respectively, confirming that the excerpt was neutral in terms of valence and neither high nor low in arousal (No. 4 signified the midpoint on the bipolar valence and arousal scales ranging from 1 to 7). Furthermore, a repeated-measures ANOVA confirmed that the elating piece (intended to bring the participants in Condition 4 to a more positive emotional state after the experiment) conveyed happiness more than any other emotion, $F(4, 76) = 93.98$, $p < .001$,

$\eta^2_G = .76$. All post hoc comparisons between happiness and other emotions were significant at the $p < .001$ level.

Discussion

The pilot experiment confirmed that the selected music excerpts conveyed the intended emotions. The experimenter-selected sad music conveyed sadness in an effective and consistent manner, and was unfamiliar to the vast majority of participants. Similarly, the neutral excerpt was rated as unfamiliar, neutral (in terms of valence), and neither high nor low in arousal. It was also confirmed that the elating excerpt conveyed happiness more than any other emotion. Thus, these excerpts were considered to be appropriate for the purposes of the actual experiment.

Experiment: Method

Participants

The participants were 120 Finnish university students from different faculties, aged 19–56 years (mean 23.7, *SD* = 4.3, 77.5% females). Ninety-seven percent of the participants were between 19 and 29 years of age. Fifty-five percent of the participants were nonmusicians, 34% had music as a hobby, and 11% were amateur or semiprofessional musicians. The participants were randomly assigned to four conditions: Participants in Condition 1 listened to unfamiliar, instrumental sad music chosen by the experimenters; participants in Condition 2 listened to neutral music chosen by the experimenters; participants in Condition 3 listened to self-selected sadness-inducing music; and participants in Condition 4 recalled an autobiographical event that had made them very sad and wrote about it (for a similar task, see Baker & Gutterfreund, 1993). There were 30 participants in every condition.

Stimuli

The participants in Conditions 1 and 2 listened to the unfamiliar sad and neutral music (respectively) described in the Pilot experiment. The participants in Condition 3 were instructed to bring approximately 8 minutes (1–2 pieces) of music that made them sad, and 8 minutes (1–2 pieces) of music that made them happy. The reasons for asking the participants to bring music that made them happy (in addition to the music that made them sad) were to conceal the true purpose of the experiment, and to use the happy music to induce a more positive emotional state after the actual experiment. Twenty-seven participants (90%) brought two sad pieces, while three participants brought one sad piece.

Measures

The participants' affective state in the beginning of the experiment was measured using the *Positive and Negative Affect Schedule* (PANAS; Watson, Clark, & Tellegen, 1988). The PANAS has 10 adjectives measuring positive affect, and 10 adjectives measuring negative affect. Prior to the experiment, the participants completed an empathy measure (*The Interpersonal Reactivity Index*; Davis, 1980). The IRI has 28 items that tap into four separate aspects of the global concept of empathy: *Fantasy, Perspective-taking, Empathic Concern*, and *Personal Distress*. The participants also filled in a short background questionnaire about their musical genre preferences, music listening habits, and musical expertise, as well as a personality measure (*The Big Five Inventory*; John & Srivastava, 1999) that was not included in the present analyses. All the measures used in the present experiment have previously been translated into Finnish.

The emotional states induced by the four different conditions were assessed using two indirect measures: a word recall task and a picture judgment task. In the word recall task, the participants had 2 minutes to write down as many adjectives as they remembered from the PANAS they completed in the beginning of the experiment (for a similar task, see, e.g., Hartig, Nyberg, Nilsson, & Gärling, 1999). Note that the participants were not instructed to memorize the PANAS adjectives. In the picture judgment task, the participants rated the emotions expressed by 25 composite pictures depicting prototypical and ambiguous facial expressions (created by Vanger, Hoenlinger, & Haken, 1998). The five prototypical facial expressions used in the present experiment expressed joy, sadness, anger, fear, and neutrality, and the 20 ambiguous expressions were computer-generated combinations of these five prototypical expressions (for details, see Vanger et al., 1998).

Procedure

The experiments were conducted individually for each participant. A special computer program was designed in MAX/MSP graphical programming environment (running on Mac OS X) to run the experiment and to collect data. In order to minimize the effect of demand characteristics, the participants were told that the purpose of the study was to investigate the effect of music listening on cognitive processing, namely memory and perception. They were not informed about the role of affect in these tasks. Participants in Condition 4 were told that they were randomly assigned to a control condition, where they would not listen to music before doing the memory and perception tasks. They were told that the effects of music listening would be compared to the control group, and that they would do the music listening task after the tasks measuring cognitive processing.

In the beginning of the experiment, the participants rated their current affective state using the 20 adjectives of the PANAS (one adjective at a time). The order of the adjectives was balanced across participants. The participants were told that the purpose of the affect measure was to control for a possible fatigue or boredom effect during the experiment. Next, participants in Conditions 1 and 2 listened to the experimenter-selected sad or neutral music, and participants in Condition 3 listened to their self-selected sadness-inducing music. They were instructed to concentrate on listening to the music, and to close their eyes if it helped them to concentrate. The duration of the music-listening part was approximately 8 minutes. Participants listened to the music pieces through studio-quality headphones, and were able to adjust the sound volume according to their own preferences.

While participants in Conditions 1, 2, and 3 listened to music, the participants in Condition 4 completed an autobiographical memory task intended to induce a sad emotional state. They received the following instructions:

> The next part is about memory and feelings. Your task is to think back to an event in your life that made you very sad. Try to remember in

> detail what happened and how you felt, and write a short description of the events in the text box below. The most important part of the task is to recall what happened in as much detail as possible. You will have 8 minutes to complete the task, after which the program will automatically move to the next part.

After the music listening task (or the autobiographical memory task), all participants completed the word recall task described in the Measures section. In order to minimize any possible negative effects this task might have on the participants' emotional states (in the case they found the task difficult; see, e.g., Hartig et al., 1999), the participants received the following instructions:

> The next part is about memory. Your task is to write down as many adjectives as you remember from the *Your current mood state* questionnaire (which you completed in the beginning of this experiment) in the text box that will appear below. You will have 2 minutes to complete this task (the time will start when you press the *OK* button). Do not start thinking about the words before pressing the button. This task does not measure intelligence or ability, and we do not expect anyone to remember all adjectives (or even close). We just want to find out how many of the adjectives you happen to remember in 2 minutes.

After 2 minutes had passed (during which the participants wrote down the adjectives they recalled), the computer program moved automatically to the picture judgment task. The 25 facial pictures were displayed in a different random order for each participant. The participants were instructed to rate the emotions each facial expression was expressing in their opinion using five rating scales (anger, fear, happiness, sadness, and neutral) ranging from 1 (*not at all angry*) to 7 (*very angry*). After the picture judgment task, the participants completed the PANAS questionnaire for the second time. Finally, participants in Conditions 1, 2, and 3 answered a few questions about the music listening task. They had to rate (on a scale from 1 to 7) how pleasant they found the music listening task (1 = *very unpleasant*, 7 = *very pleasant*), and how much they liked the music piece(s) they listened to (1 = *not at all*, 7 = *very much*). They were also asked to describe (by writing into a text box) what kinds of thoughts or impressions went through their minds during the music listening. Similarly, participants in Condition 4 were asked to rate how pleasant they found the autobiographical memory task. Participants in Conditions 1 and 2 finished the experiment after these final questions, but participants in Conditions 3 and 4 listened to happy or elating music for approximately 8 minutes before debriefing in order to ensure that they would be in a more positive emotional state by the end of the experiment. Participants in Condition 3 listened to their self-selected, happiness-inducing music, while participants in Condition 4 listened to the experimenter-selected elating music. After the experiment, all participants were fully debriefed and given a free movie ticket (value 7–14 €) in return for their participation.

Results

Memory Effects

The words were coded as correctly recalled if the root meaning of the word was correct. For example the words *kärttyisä* and *kärttyinen* (both mean *irritable*) were regarded as equivalent, as were the words *pelokas* (*fearful*) and *peloissaan* (*afraid*). The mean numbers of correctly recalled positive and negative words (for each condition) are displayed in Figure 1. In order to construct an aggregate measure combining the memory for both positive and negative words (as the number of words remembered varied considerably between participants), the number of correctly recalled negative words was deducted from the number of correctly recalled positive words. As it was expected that the differences between the neutral and sad conditions would be subtle (in comparison to previously reported differences between happy and sad affect inductions), two planned comparisons of means were carried out to test the competing hypotheses 1A [+1 −3 + 1 +1] and 1B [−1 −1 + 1 +1]. Hypothesis 1A predicted that participants in Conditions 1 (unfamiliar sad music), 3 (self-selected sad music), and 4 (sad recall) would display an affect-incongruent memory bias (i.e., more positive words remembered in relation to negative words), whereas participants in Condition 2 (neutral music) would not display a memory bias. In comparison, Hypothesis 1B predicted that only participants in Conditions 3 and 4 would display an affect-incongruent memory bias. The planned comparisons revealed that participants in Conditions 3 and 4 recalled more positive words in relation to negative words than participants in Conditions 1 and 2, giving support to Hypothesis 1B; $F(1, 118) = 7.92$, $p < .01$, $\eta^2 = .06$. The analysis concerning Hypothesis 1A did not reach statistical significance.

The recalled positive and negative words were further investigated within conditions using two-tailed, paired *t* tests. In Conditions 1 and 2, there was no difference between the number of correctly recalled positive and negative words; $t(29) = 0.41$, *ns*, and $t(29) = 0.32$, *ns*, respectively. In Condition 3, the difference between positive and negative words was not quite significant; $t(29) = 1.81$, $p = .080$, while in Condition 4, the difference was clearly significant; $t(29) = 4.80$, $p < .001$. The mean numbers of correctly recalled positive and negative words (in each condition) are illustrated in Figure 1.

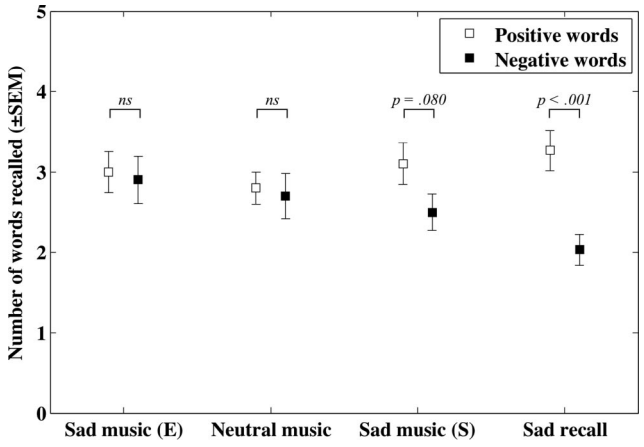


Figure 1. The mean number (± standard error of the mean) of positive and negative adjectives recalled by the participants in the different conditions (E = experimenter-selected sad music, S = self-selected sad music). The within-condition differences between the numbers of recalled positive and negative words were calculated using paired, 2-tailed *t* tests.

Judgment Effects

To eliminate the effect of possible individual differences in scale use, the raw emotion ratings for the facial expression pictures were standardized within subjects using individual *z*-score transformations. The *z*-scores were calculated using all emotion ratings of each participant. According to the affect-congruency theory (e.g., Bower, 1981), people experiencing a sad emotional state should give lower happiness ratings and higher sadness ratings to emotional stimuli. To investigate whether the mean happiness ratings (calculated from each participant's happiness ratings for all facial expressions) differed between certain conditions, planned comparisons were carried out to test hypotheses 1A [−1 + 3 −1 −1] and 1B [+1 + 1 −1 −1]. The results again supported Hypothesis 1B, $F(1, 118) = 6.52$, $p < .05$, $\eta^2 = .05$, as the analysis regarding Hypothesis 1A did not quite reach statistical significance ($p = .054$). The analyses confirmed that participants in Conditions 3 (self-selected sad music) and 4 (sad recall) gave lower happiness ratings to facial expressions than participants in Conditions 1 (unfamiliar sad music) and 2 (neutral music), thus displaying an affect-congruent judgment bias. Similar analyses were also conducted for mean sadness ratings. Both analyses failed to reach statistical significance, but the trend favored Hypothesis 1A [+1 −3 + 1 +1]: $F(1, 118) = 2.88$, $p = .092$.

Mood Prior to the Experiment

In order to confirm that the memory and judgment biases reflected the hypothesized changes in the participants' emotional states, 1-way ANOVAs were carried out to analyze the PANAS ratings collected in the beginning of the experiment. There was no main effect of condition in the PANAS ratings, confirming that there were no existing mood differences between the groups prior to the music listening or autobiographical recall. $F(3, 116) = 0.23$, *ns* for positive affect, and $F(3, 116) = 0.25$, *ns* for negative affect.

Individual Differences

To investigate Hypothesis 2, according to which highly empathic people may be more susceptible than others to emotional states induced by unfamiliar music, we had a closer look at the mean sadness ratings for facial expressions—the only variable suggesting a trend toward Hypothesis 1A. As trait empathy was hypothesized to play a key role in the process, we proceeded by investigating whether trait empathy and/or its subscales were associated with affect-congruent biases in the emotion ratings. Table 1 displays the correlations between global Empathy (and its subscales) and the participants' mean sadness ratings. Global Empathy and the Empathy subscale *Empathic Concern* correlated significantly with the mean sadness ratings in Condition 1 ($r = .63$, $p < .001$; $r = .48$, $p < .01$), and the Empathy subscale *Fantasy* correlated with mean sadness ratings both in Condition 1 ($r = .54$, $p < .01$) and Condition 3 ($r = .49$, $p < .01$). There were no statistically significant correlations in the other two conditions. These results suggest that empathy is associated with the susceptibility to musically induced emotional states, as trait empathy was connected to affect-congruent judgment biases only in the conditions where sad music—experimenter-chosen sad music in particular—was used to induce an emotional state.

Table 1
Correlations Between Empathy (and its Subscales) and the Participants' Mean Sadness Ratings for Facial Pictures (Grouped by Condition)

	Mean sadness ratings for facial pictures			
	Sad music (E)	Neutral music	Sad music (S)	Sad recall
Empathy (global)	.63***	−.08	.34	.25
Fantasy	.54**	−.21	.49**	.09
Empathic concern	.48**	.25	.31	.25
Perspective-taking	.27	−.12	−.21	.26
Empathic distress	.07	−.01	.13	.00

Note. E = Experimenter-selected sad music; S = Self-selected sad music.
** $p < .01$. *** $p < .001$.

This finding is further illustrated by Figure 2, which displays the combined happiness and sadness ratings of participants scoring high and low (median split) in global Empathy. Affect models typically characterize happiness and sadness as diametric opposites (see, e.g., Russell, 1980; Watson & Tellegen, 1985). As there were an equal number of prototypical and ambiguous (mixed) facial pictures expressing sadness and happiness (and as all emotion ratings were standardized within participants), the participants' mean sadness ratings were deducted from their happiness ratings in order to investigate a general judgment bias on the sad–happy continuum. Thus, in Figure 2, zero means no judgment bias, positive values signify a judgment bias toward happiness, and negative values signify a judgment bias toward sadness. In order to test for differences between high-empathy and low-empathy participants (median split) within conditions, two-tailed *t* tests were used to analyze the individual judgment biases on the sad–happy continuum. In Condition 1 (unfamiliar sad music), the difference between high-empathy and low-empathy participants was statistically significant, $t(28) = -2.29$, $p < .05$, indicating that the high-empathy participants were displaying a stronger judgment bias toward sadness. In comparison, there were no differences between high- and low-empathy participants in Conditions 2, 3, or 4; $t(28) = 0.94$, $t(28) = -0.23$, and $t(28) = 0.78$ (respectively), all *p*s = *ns*.

Next, planned comparisons investigating the combined happiness and sadness ratings were carried out separately for participants scoring high and low in global Empathy (median split). For the high-empathy participants, the results favored Hypothesis 1A [−1 + 3 −1 −1], $F(1, 57) = 6.68$, $p < .05$, $\eta^2 = .10$, while the results regarding the low-empathy participants gave support to Hypothesis 1B [+1 + 1 −1 −1], $F(1, 59) = 5.86$, $p < .05$, $\eta^2 = .09$. In other words, the high-empathy participants in Condition 1 (unfamiliar sad music) had a judgment bias toward sadness that was significantly different from Condition 2 (neutral music)—similarly to the participants in Conditions 3 and 4. In comparison, the low-empathy participants in Condition 1 did not differ from the neutral condition in terms of judgment biases.

As most of the participants were female (77.5%), we also investigated whether gender contributed to the emotional states induced by music or autobiographical recall. Two-way ANOVAs revealed no main effects of gender on the word recall task or the

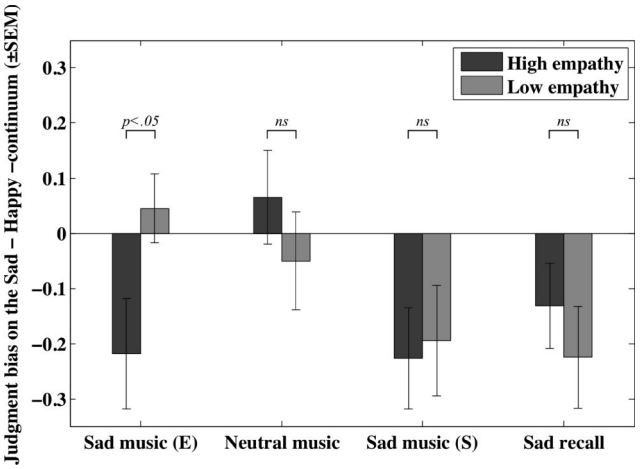


Figure 2. The mean judgment biases (± standard error of the mean) in the combined sadness and happiness ratings (for facial expressions) of participants scoring high and low in trait empathy (median split), grouped by condition (E = experimenter-selected sad music, S = self-selected sad music). Zero equals no judgment bias, positive values signify a judgment bias toward happiness, and negative values signify a judgment bias toward sadness. The within-condition differences between high- and low-empathy participants were calculated using 2-tailed *t* tests.

picture judgment task, and no interaction (condition × gender) effects. Regarding the personality traits associated with a susceptibility to musically induced emotional states, 1-way ANOVAs revealed main effects of gender in global Empathy, $F(1, 118) = 8.06$, $p < .01$, $\eta^2 = .06$, and its subscale Empathic concern, $F(1, 118) = 7.53$, $p < .01$, $\eta^2 = .06$. For females, the mean Empathy score was 99.98 ($SD = 8.47$), and for males 94.15 ($SD = 12.13$). However, there was no main effect of gender on the subscale Fantasy, $F(1, 118) = 3.65$, $p = .059$. These findings are in line with previous research that has reported gender-related differences in self-reported empathy, while physiological, gestural, and facial measures of empathy have revealed nonexistent gender differences (for a meta-analysis, see Eisenberg & Lennon, 1983).

Free Descriptions of Thoughts That Occurred During Music Listening

After completing the indirect (and direct) measures of emotional states, the participants were asked to describe what kinds of thoughts or impressions went through their minds while they were listening to the music. The free descriptions of participants in Conditions 1 and 3 were coded using classical content analysis (see, e.g., Ryan & Bernard, 2000), with the aim to construct classifying themes that would account for the broad, common constituents of the participants' thoughts and impressions. The emotion-induction mechanisms proposed by Juslin and Västfjäll (2008) were used as a framework for the coded themes, although other themes also emerged from the data. Each free response was coded according to the predominant theme. The data were first coded by one of the experimenters, and then verified by the other experimenter using the same criteria (Cohen's $\kappa = .77$). The few mismatching cases were discussed and adjusted accordingly. The found themes were: 1. *Sad imagery* (e.g., "The piece was quite melancholic and made me think of the ending of a sad movie. Maybe someone had died or gone away and left someone else missing him/her"), 2. *Sad memories* (e.g., "I thought a lot about my grandmother who passed away, and how I wouldn't want to lose anyone else"), 3. *Other imagery* (e.g., "The piece evoked impressions of nature; storm, water, wind, and swaying trees"), 4. *Unspecified memories* (e.g., "The pieces brought back memories from the time when I used to listen to these pieces a lot"), 5. *Sadness mentioned in another context* (e.g., "The piece sounded sad", or "I felt a little melancholic"), and 6. *Other* (e.g., " I felt sleepy", or "I thought about the artist's baby"). The distribution of the different themes in Conditions 1 and 3 is summarized in Table 2. One participant in Condition 3 did not provide any free descriptions, and has thus been omitted from the table.

As can be seen from Table 2, here were notable differences between the free descriptions of the participants in the two conditions: in Condition 1, seven participants (23%) reported thinking about sad imagery and six reported thinking about other imagery, whereas in Condition 3, the number of participants with the same themes were two and one. In contrast, 14 participants (47%) in Condition 3 thought about sad personal memories and four thought about unspecified personal memories, while in Condition 1, only one participant thought about sad memories and two thought about unspecified memories. In addition, one participant in Condition 1 and two participants in Condition 3 reported crying during the music listening. None of these three participants reported thinking about personal memories during the music listening.

Pleasantness of Music Listening and Autobiographical Recall

The mean pleasantness ratings for the music listening tasks and the autobiographical recall task were 5.7 (unfamiliar sad music), 5.2 (neutral music), 5.2 (self-selected sad music), and 3.6 (sad recall). The ratings were made on a scale from 1 (*very unpleasant*) to 7 (*very pleasant*). A 1-way ANOVA yielded a main effect of condition, $F(3, 116) = 13.88$, $p < .001$, $\eta^2 = .26$, and post hoc analysis (Tukey) revealed that the pleasantness ratings in Condition 4 (sad recall) differed significantly from all other conditions ($p < .001$). It is interesting to note that listening to the experimenter-selected sad music was found to be more pleasant than listening to the neutral music or self-selected sad music, although these differences were statistically nonsignificant.

Table 2
The Distribution of the Themes Found in the Free Descriptions (About Thoughts and Impressions That Occurred During Music Listening) of Participants Who Listened to Experimenter-Selected, Unfamiliar Sad Music (E), and Participants Who Listened to Self-Selected Sad Music (S)

	Sad music (E)	Sad music (S)
Sad imagery	7 (23%)	2 (7%)
Sad memories	1 (3%)	14 (47%)
Other imagery	6 (20%)	1 (3%)
Unspecified memories	2 (7%)	4 (13%)
Sadness mentioned (other)	7 (23%)	6 (20%)
Other	7 (23%)	2 (7%)

Liking Ratings for Music Pieces

Finally, we investigated the liking ratings given to the music pieces. The mean ratings were 5.3 (unfamiliar sad music), 4.6 (neutral music), and 6.4 (self-selected sad music). Neutral music was the least liked, while self-selected sad music was the most liked. A 1-way ANOVA yielded a main effect of condition, $F(2, 87) = 15.48$, $p < .001$, $\eta^2 = .26$, and post hoc analyses (Tukey) revealed that the familiar, self-selected sad music was liked significantly more than unfamiliar, experimenter-selected sad music ($p < .01$) or neutral music ($p < .001$).

Discussion

This study aimed to investigate whether sad music could induce genuine sadness in listeners. Two indirect measures of experienced affect—a word recall task and a picture judgment task—demonstrated that self-selected sadness-inducing music (Condition 3) could indeed induce memory and judgment effects that were significantly different from the neutral condition—despite the fact that the self-selected, familiar music was loved by the participants. However, experimenter-selected, unfamiliar sad music (Condition 1) managed to induce a sad emotional state only in empathic participants.

Participants in Conditions 3 (self-selected sad music) and 4 (autobiographical recall) displayed affect-incongruent biases in the word recall task. Although the associative network theory of affect (Bower, 1981) postulates that affective states should lead to affect-congruent memory effects, it has been demonstrated that affect-incongruent memory effects emerge when participants are unaware of the relevance of their moods and emotions to the experiment (Parrot & Sabini, 1990), or when the recalled material is only mildly pleasant or unpleasant (Rinck, Glowalla, & Schneider, 1992). Both specifications were met by the present experiment, as participants were not informed about the relevance of their emotional states to the experiment, and most of the PANAS adjectives in the word recall task were not strongly pleasant or unpleasant. The memory bias was somewhat stronger in Condition 4 (compared to Condition 3), which may result from differences in the two affect-induction tasks, as well as from the type of adjectives used for the word recall task. As the autobiographical recall task was experienced as significantly more unpleasant than the listening task (with self-selected sadness-inducing music), it is possible that the recalled pleasant and unpleasant adjectives reflected the difference in the pleasantness of the two tasks—in addition to the induced sad emotional state. This may also partly explain the lack of mood-incongruent memory effects in Condition 1, as listening to the unfamiliar sad music was found to be even more pleasant than listening to the neutral music or self-selected sad music.

The judgment task (where participants evaluated emotions expressed by facial pictures) revealed strong, affect-congruent judgment biases in Conditions 3 and 4. As this task measured specifically the judgment bias in perceived happiness and sadness, it is plausible that the task reflected more clearly the induced (sad) emotional states. The judgment bias in sadness ratings was also the only variable suggesting a trend toward Hypothesis A, according to which experimenter-selected, unfamiliar sad music (Condition 1) also would be able to induce genuine sadness in participants. However, closer examination revealed that the experimenter-selected sad music was able to successfully induce a sad emotional state only in participants scoring high in trait empathy. Highly empathic participants in Condition 1 displayed a judgment bias that was significantly different from the neutral condition, whereas participants scoring low in empathy did not display a judgment bias. These individual differences in the susceptibility to music-induced sadness have implications not only for the study of music and emotions, but also for studies aiming to affect the participants' emotions or moods through music listening. As the emotional responsiveness to unfamiliar music seems to be strongly associated with trait empathy, this limits the effect sizes and generalizability of findings in studies utilizing such stimuli.

As the vast majority of the participants in Condition 1 did not report thinking about personal memories during the music listening, the significant connection between empathy and sadness induced by unfamiliar sad music suggests that real sadness can be experienced through emotional contagion—a subcomponent of empathy as well as one of the emotion induction mechanisms suggested by Juslin and Västfjäll (2008). It may also be that unfamiliar, instrumental sad music is able to induce sadness through visual imagery (see Juslin & Västfjäll, 2008), as 23% of the participants in Condition 1 described thinking about sad imagery during the music listening.

In comparison, self-selected sad music was strongly associated with autobiographical memories, as 47% of the participants in Condition 3 reported thinking about sad memories, and 13% reported thinking about unspecified memories during the music listening. This suggests that the mechanism episodic memories (see Juslin & Västfjäll, 2008) plays an important role in sadness induced by music that has personal relevance. It is also worth noting that the content of lyrics may have contributed to the induced emotional states, as 91.5% of the self-selected sad music contained lyrics. This possibility is rendered probable by the findings of previous studies (e.g., Ali & Peynircioylu, 2006; Brattico et al., 2011) that suggest lyrics may be important for music-induced sadness. However, there may be other mechanisms as well—such as emotional contagion—that contribute to sadness induced by personally relevant, familiar music, since Fantasy—the tendency to imaginatively transpose oneself into the feelings and actions of fictitious characters (Davis, 1980) —was related to higher mean sadness ratings also in Condition 3.

Global Empathy and Fantasy have previously been associated with the enjoyment of sad music (Vuoskoski et al., in press; Garrido & Schubert, 2011). As these two personality attributes were related to the susceptibility to music-induced sadness in the present study, it could be speculated that the intensity of music-induced sadness and the enjoyment of sad music are connected. If this is indeed the case, it opens new, intriguing directions for future studies seeking to explain the mechanisms behind the pleasure generated by sad music (or music-induced sadness; cf., Huron, 2011). These findings may also have implications for music therapy applications, as music therapy is effectively used in the treatment of depression (see, e.g., Erkkilä et al., 2011), for example.

In summary, this study demonstrated that sad music could indeed induce sadness-related effects on memory and judgment. However, this effect depends—to some extent—on the music's relevance to the listener, as well as on the personality attributes of the listener. Most noteworthy is the fact that this study demon-

strated that genuine sadness could be induced by unfamiliar sad music, without any apparent autobiographical content—challenging the longstanding claims of the musical "cognitivists" (e.g., Kivy, 1989, 1990). Trait empathy contributed to the susceptibility to sadness induced by unfamiliar, instrumental music, while autobiographical memories contributed to sadness induced by self-selected music. Listening to sadness-inducing music was found to be significantly more pleasant than the recollection of sad autobiographical memories, suggesting that while sad music is able to induce similar effects on emotion-related memory and judgment as autobiographically induced sadness, the subjective experience of music-induced sadness is intrinsically more pleasurable.

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